

Shivani Chaudhary

B.Tech. Graduate (2023)

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Work Experience

Software Engineer | Balbix, Cybersecurity Company

(May'23- Present)

• Agent Workflow & Event-Driven Systems

- Developed configurable **third-party workflow agents** with custom triggers and pluggable step types, supporting **automated** event-driven execution as well as manual workflow invocation for **third-party onboarding**
- Architected **bi-directional synchronization** between tier settings and predefined workflows, eliminating configuration drift and ensuring consistency across UI and backend execution

• AI-powered Chatbot

- Built an **AI-powered security chatbot** that combined enterprise knowledge with **live customer security data** to deliver personalized, **context-aware responses** through natural-language interactions
- Implemented the end-to-end conversational AI platform using **RAG pipelines**, and **MCP-based tool calling**, enabling **real-time streaming** and **multi-turn conversations** over live security data

• Fourth-Party Pipeline, API Optimization & AI Insights

- Developed a **fourth-party discovery pipeline** by extending a distributed **ECS-based unified engine**, integrated with **Airflow** and **Snowflake** for scalable risk data enrichment across services
- Engineered high-performance **Snowflake-backed APIs**, improving query latency and eliminating data inconsistencies, enabling **AI-driven summaries** across analytics and user-facing systems at scale

• API Development

- Developed Python **Flask** APIs serving **AI/ML** model data for vulnerability risk scoring and asset behavior analysis, handling **30–50 RPS** with reliable performance
- Built **100+ GraphQL** queries using **Apollo Client**, **Node.js**, and **Express.js** with a **structured API framework** and robust **error handling**, cutting **UI disruptions by 30%**

• Data Ingestion & Integration

- Engineered **connectors** to ingest asset data from Tenable, Qualys, CrowdStrike, Microsoft Defender, and ServiceNow CMDB, processing **5–15 GB** of data per customer weekly
- **Sanitized** data from cybersecurity tools using schema validation, type normalization, deduplication, and JSON standardization, improving data quality for **20–30%**
- Optimized data ingestion workflows, achieving a **15%** efficiency increase and enhancing the data pipeline's speed

• Automated Data Export Solutions

- Automated weekly data export pipelines via Python scripts, extracting data from **Elasticsearch** and delivering JSON outputs to **Amazon S3** and via **Postman API**, reducing manual effort by **90%**
- Implemented secure, scheduled API-based transfers using **Postman** and **cron**, ensuring reliable customer data delivery with **99%** data accuracy and zero manual intervention

Key Projects

• AI-Powered Knowledge Base Search & Enrichment 🗄️ (GitHub) (Self Project)

(Oct'25)

- Implemented an end-to-end **RAG pipeline** with **ingestion**, **chunking**, **SentenceTransformer embeddings**, **FAISS indexing**, SQLite storage, retrieval logic, and incremental updates
- Built FastAPI-based semantic search and Q&A using Gemini, integrated with a Streamlit UI

• Movie Recommendation System 🎬 (GitHub) (Self Project)

(July'22)

- Implemented backend part of Movie Recommendation and algorithms by using **Spyder** and **Jupyter Notebook**
- Developed and optimized a **hybrid recommendation system** in **Python** using **machine learning** algorithms such as **KNN**, combining **content-based** and **collaborative filtering** to improve accuracy and relevance

Technical Skills

- **Programming Languages:** C, C++, Python, Golang, Java, JavaScript, TypeScript, SQL
- **Technologies & Tools:** React.js, Next.js, Node.js, Express.js, Kafka, Docker, Kubernetes, Git, Gitlab, LaTeX, MySQL, PostgreSQL, Elasticsearch, HTML5, CSS, Tailwind, Material-UI, Vite, GraphQL, Postman, phpMyAdmin, n8n, Jenkins, CI/CD Pipeline, AWS, SNS, SQS, Lambda, Airflow, Snowflake, RabbitMQ, Cursor, Jest, Redis, LLMs, RAG Pipeline, MCP server, DynamoDb, Distributed System